



**National Institute of Technology Rourkela**  
**Department of Computer Science Engineering**  
Deep Learning (CS6510)  
End-Semester Examination  
B.Tech 4<sup>th</sup> year/PhD/M.Tech(R)

Duration: 3 hrs

Summer 2026, 8<sup>th</sup> Semester

Total marks: 50

**Attempt all questions;**

⟨CO-x: Course Outcome level⟩; ⟨BL-x: Bloom's taxonomy level⟩;

*\*All parts of the same question must be answered together*

1. (a) Distinguish between supervised and unsupervised learning? Given a dataset  $D = \{(x_i, y_i) : i \in N\}$ , explain the above two learning schemes. ⟨CO-1,2⟩ ⟨BL-2, 3⟩ [5]  
(b) Using a single-layer perceptron, simulate the following combinatorial circuit elements. Find the Weight vectors ( $W_i$ ) and Bias ( $\beta$ ) for each of the following. The circuit has two inputs  $x_1, x_2$ . ⟨CO-1,2⟩ ⟨BL-2, 3⟩ [5]
  - i) OR gate
  - ii) AND gate
  - iii) OR gate followed by an AND gate with the second gate (OR) receiving two inputs: first as the output from the AND gate, and second from the source input  $x_2$ .
2. (a) Define Gradient Descent (GD) in the context of training an ANN. Compare and contrast all the variants of GD with appropriate mathematical formulations. ⟨CO-3⟩ ⟨BL-3⟩ [5]  
(b) Explain the building of a confusion matrix for a binary classifier? Define and distinguish between the individual components:  $TP, FP, FN, TN$  of the matrix. ⟨CO-2,3⟩ ⟨BL-5⟩ [5]
3. (a) Explain the bias-variance trade-off with an appropriate diagram? Derive the equation for predicted error in terms of bias and variance. ⟨CO-3⟩ ⟨BL-3⟩ [5]  
(b) Define the following terms with appropriate formulations and diagrams if necessary: precision ( $p$ ); recall ( $r$ ); F1-score; sensitivity; specificity, ROC curve. ⟨CO-3⟩ ⟨BL-3⟩ [5]
4. (a) What are the disadvantages of using a fixed learning rate  $\eta$ ? Compare and contrast various optimizers for tuning the time-dependent learning rate  $\eta_t$  with appropriate mathematical formulations. ⟨CO-3⟩ ⟨BL-3⟩ [5]  
(b) Assertion: Given a DNN (Deep NN) of multiple layers. Suppose the core computation happens in only 10% of the layers while 90% of the layers are deemed as redundant. ⟨CO-3,4⟩ ⟨BL-4,5⟩ [5]

Conclusion(s) (May be True/False): The ANN can be made efficient by using only the 10% of the active layers. The backpropagation will not have a major impact on the remaining layers.

Reason(s): —

5. (a) What is the role of regularizers in loss functions? Explain in detail the working of L1 and L2 regularizers with their appropriate formulation.  $\langle \text{CO-5} \rangle \langle \text{BL-4} \rangle$  [5]
- (b) Given an input image of dimensions  $227 \times 227 \times 3$ ; For Conv1 layer:  $K = 96, F = 11, S = 4, P = 0$ . Find  $W_2, H_2$  and the number of parameters involved.  $\langle \text{CO-5} \rangle \langle \text{BL-5} \rangle$  [5]

*N.B: Make your own logical assumptions (if required) to answer wherever necessary.*